



UNIVERSITY OF
WATERLOO

What metrics do networked applications care about?

CS 456/656, Fall 2026

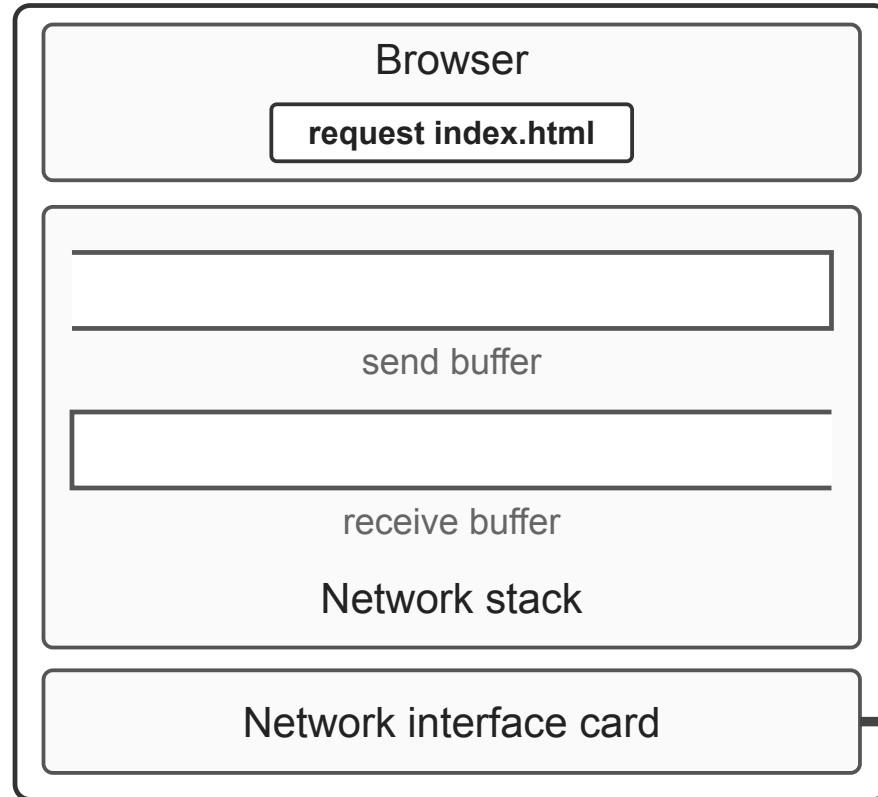
Mina Tahmasbi Arashloo

Outline

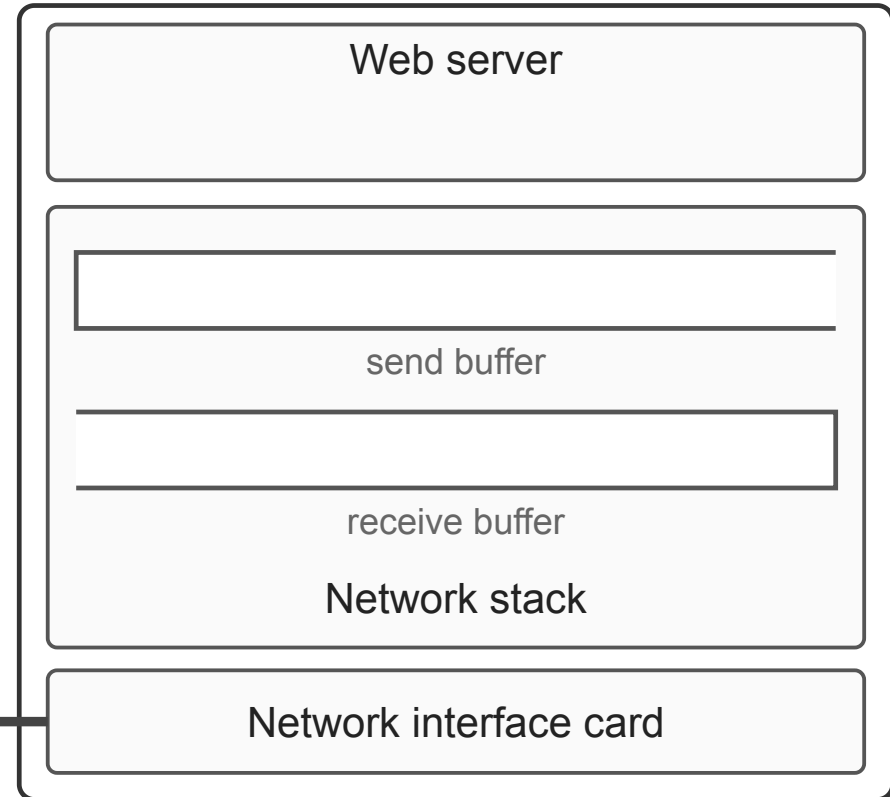
- **Where we were**
- Video streaming on a perfect network
- Video streaming on a real network
- What each application wants from the network
- Takeaways

Last time: we built a web browser

A: client

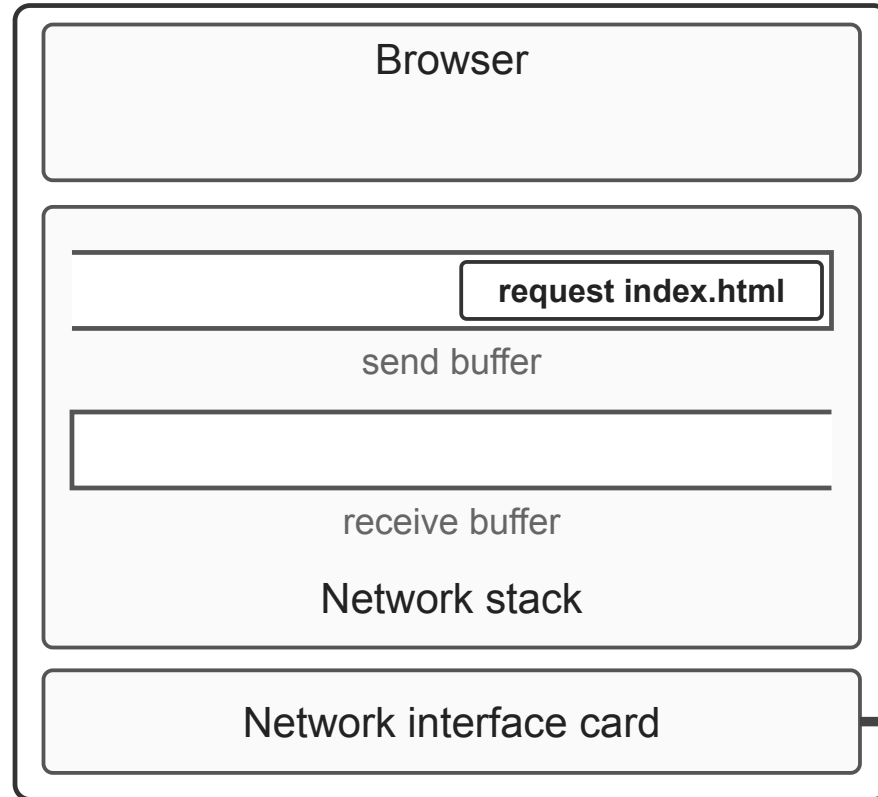


B: server

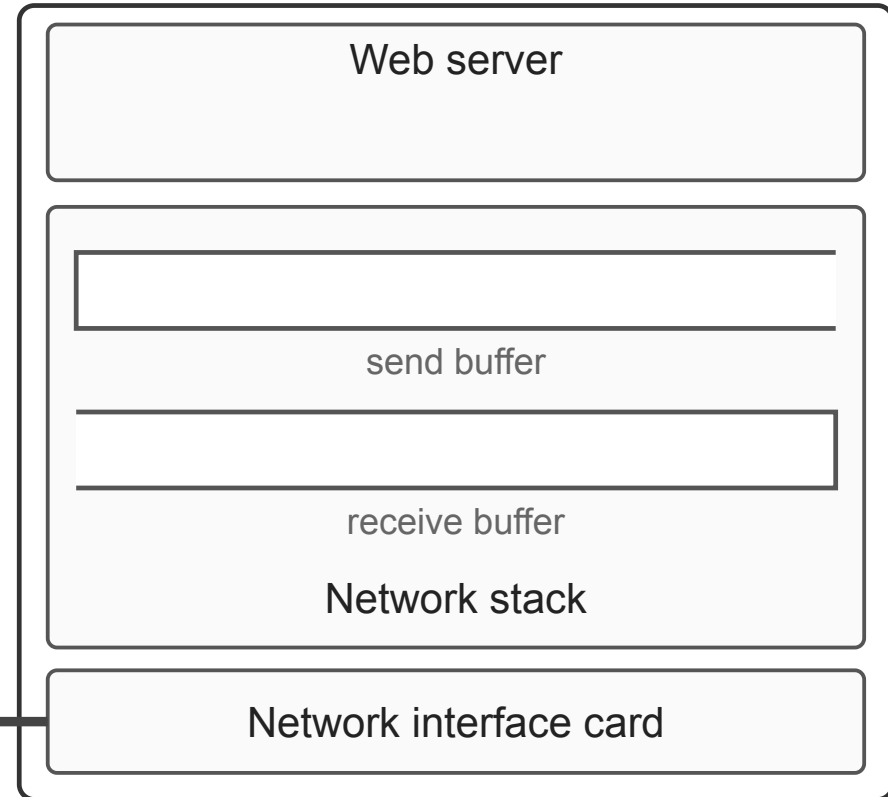


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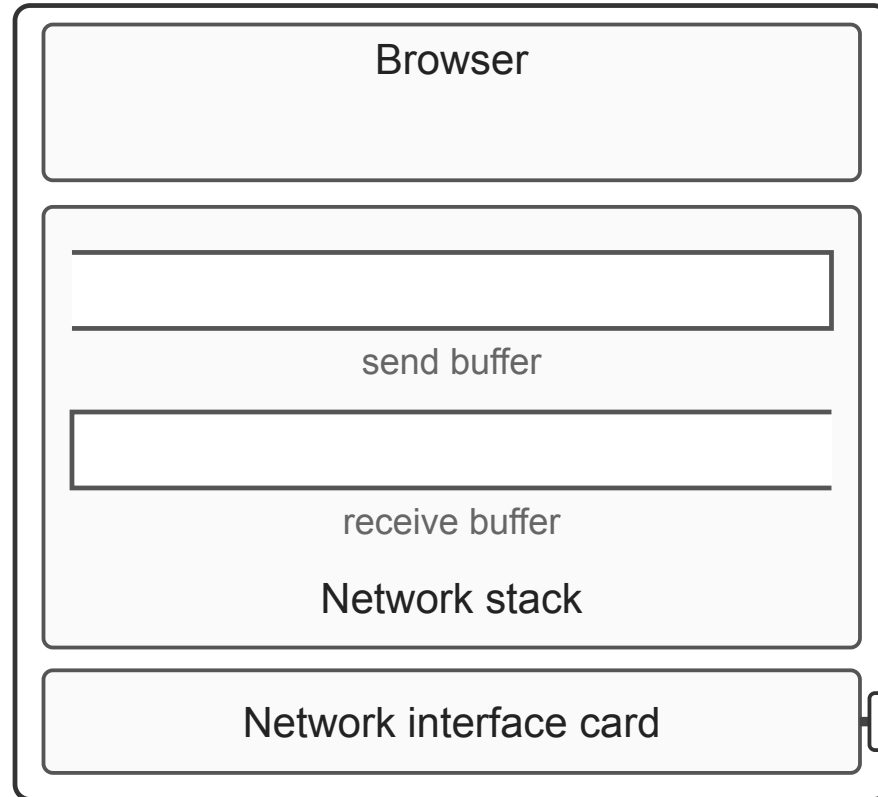


B: server

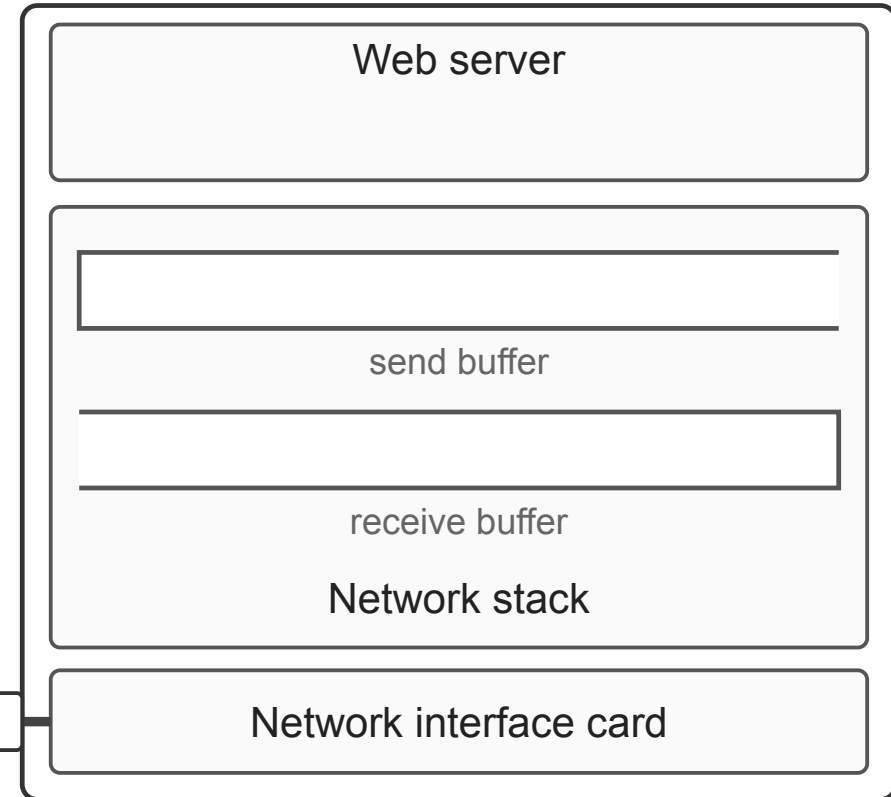


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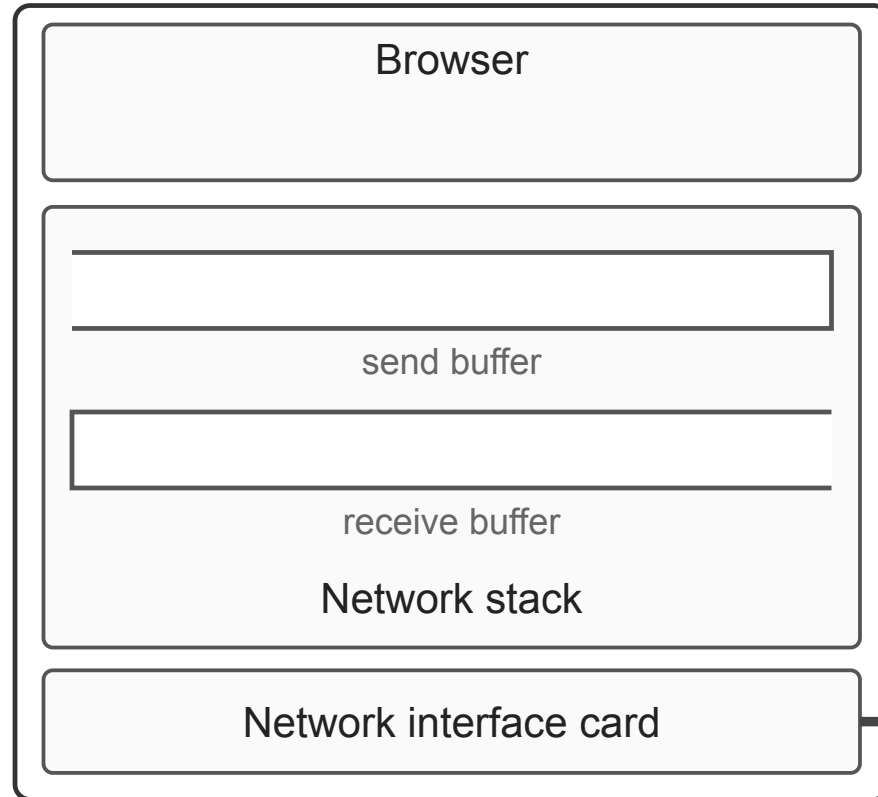
B: server



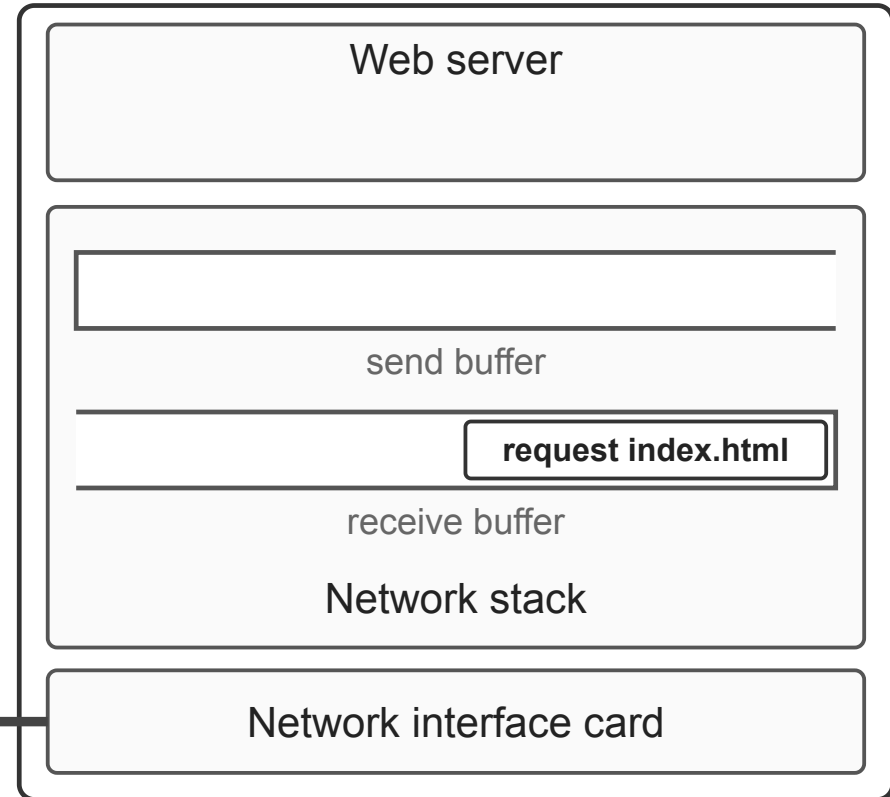
request index.html

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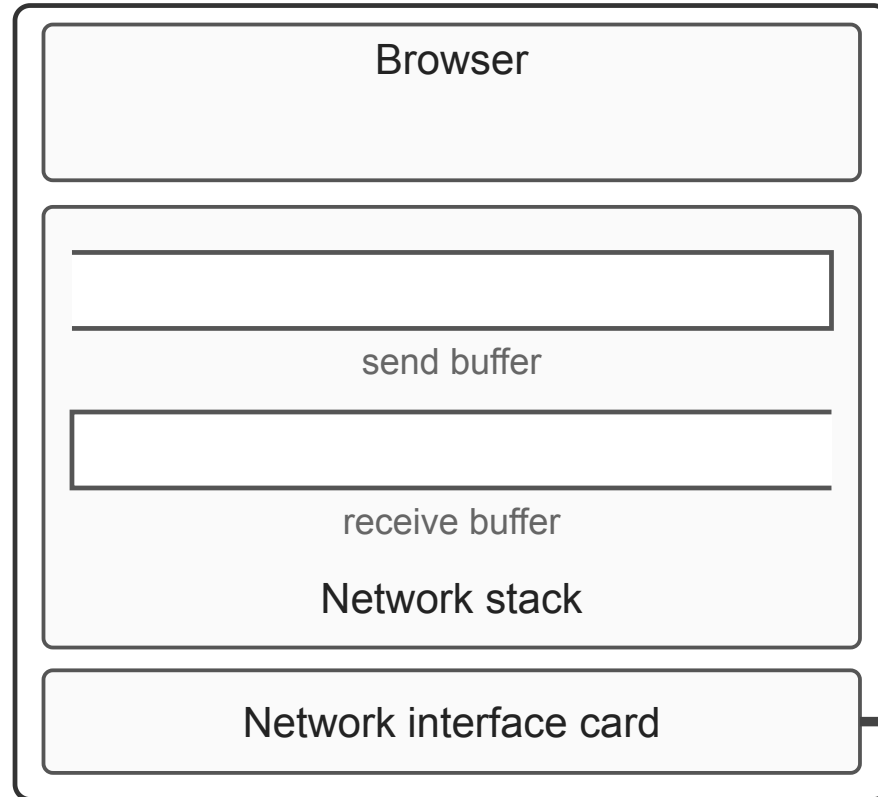


B: server

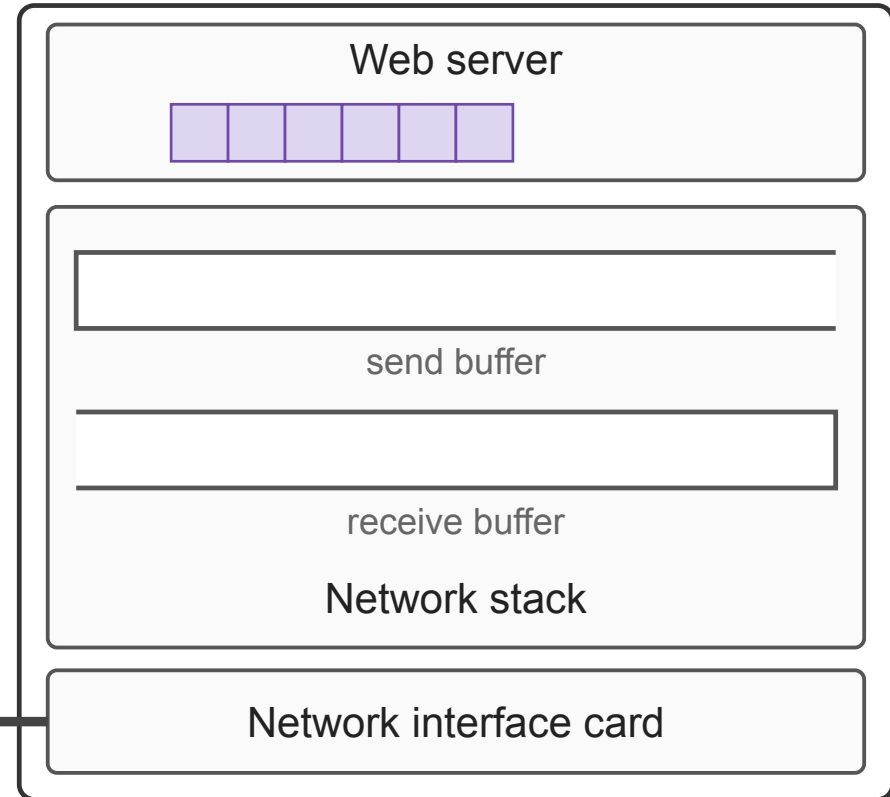


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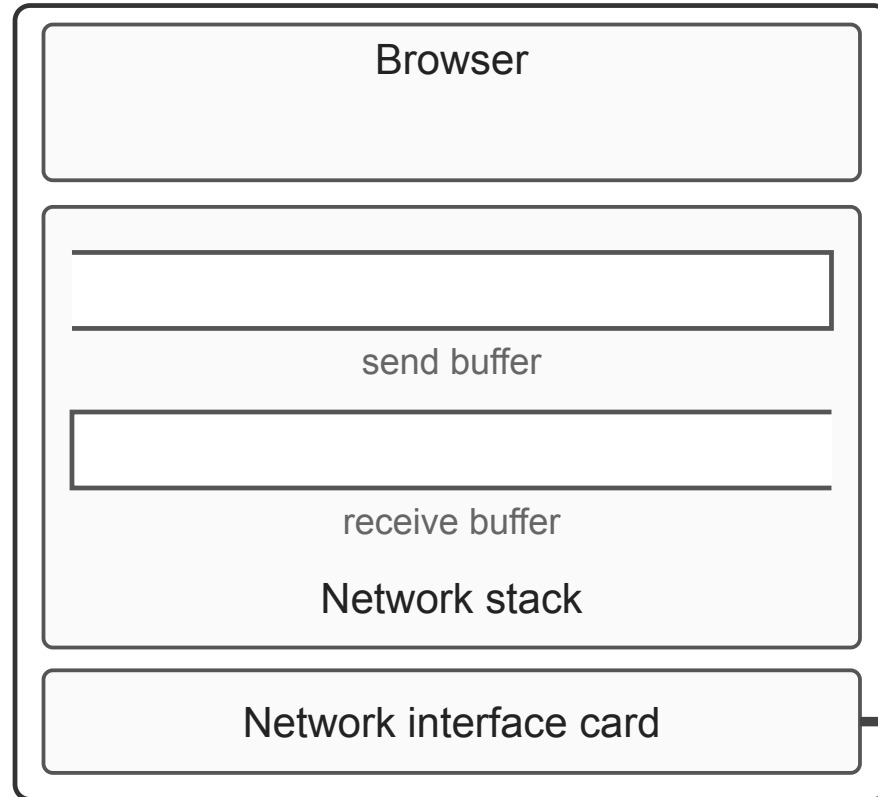


B: server

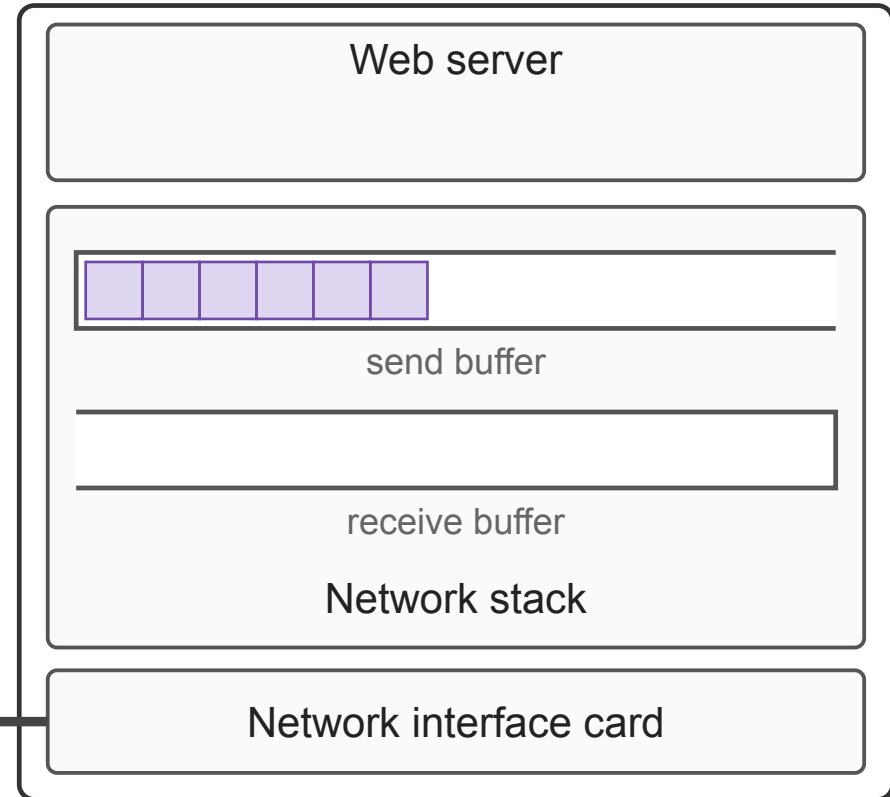


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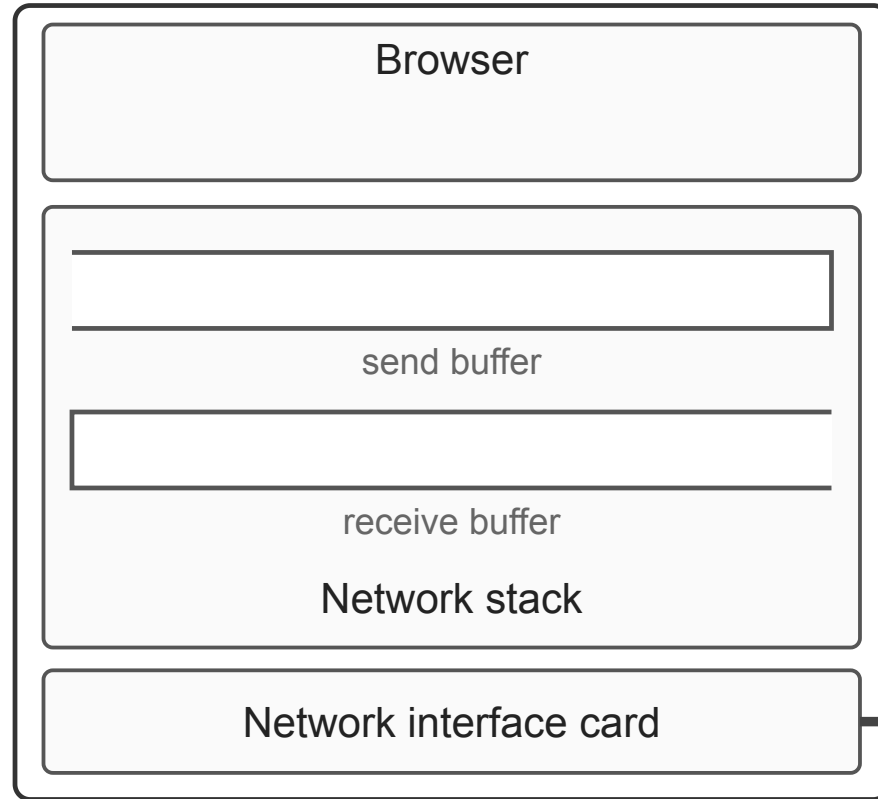


B: server

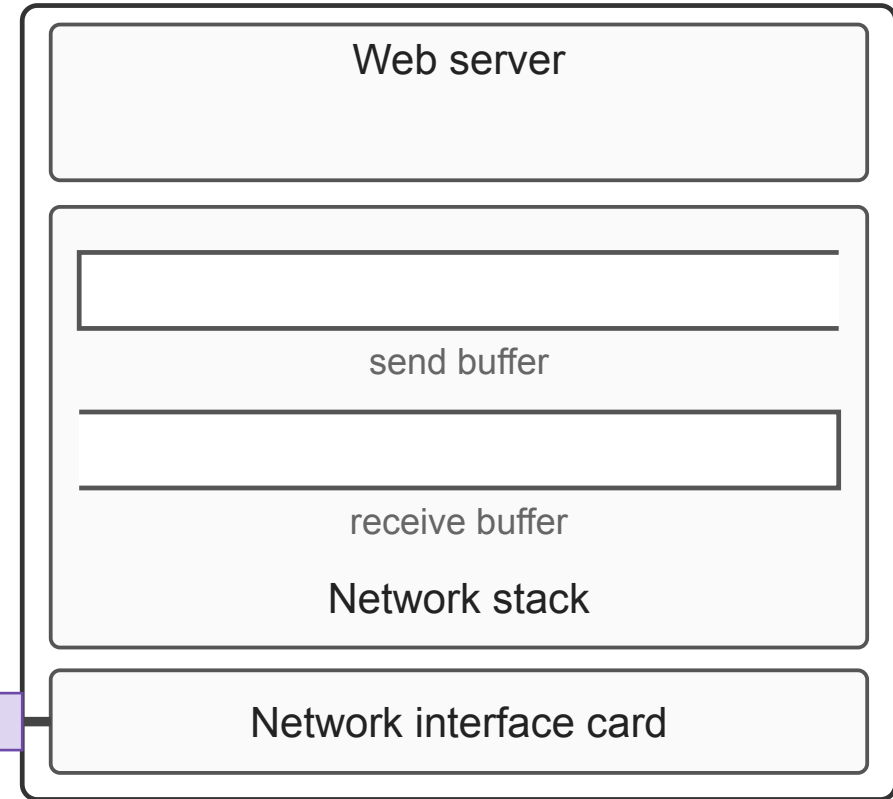


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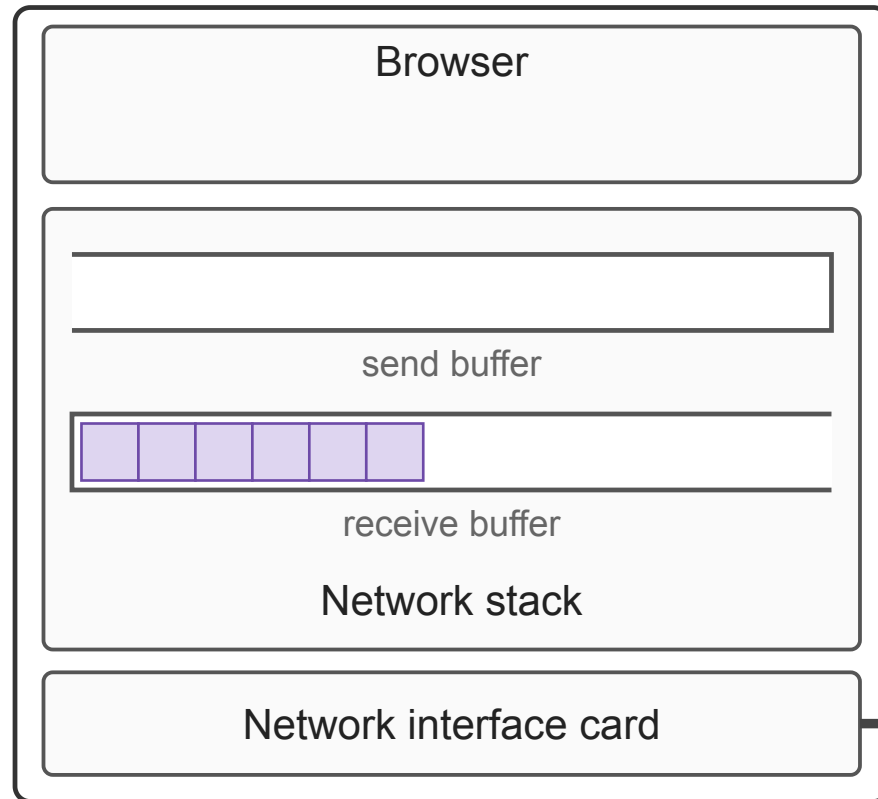


B: server

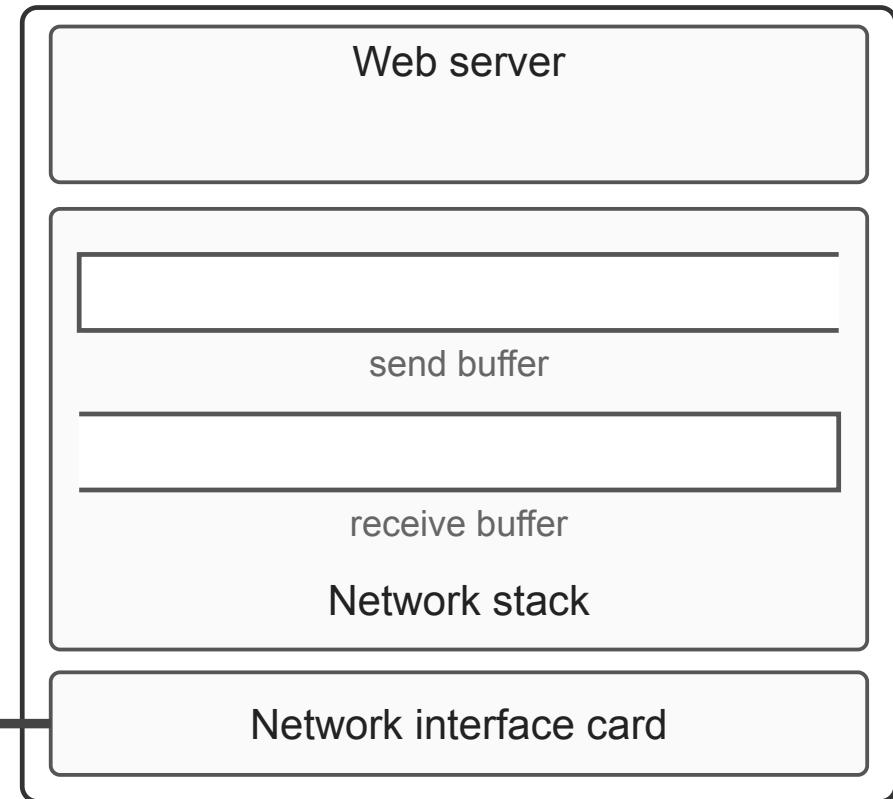


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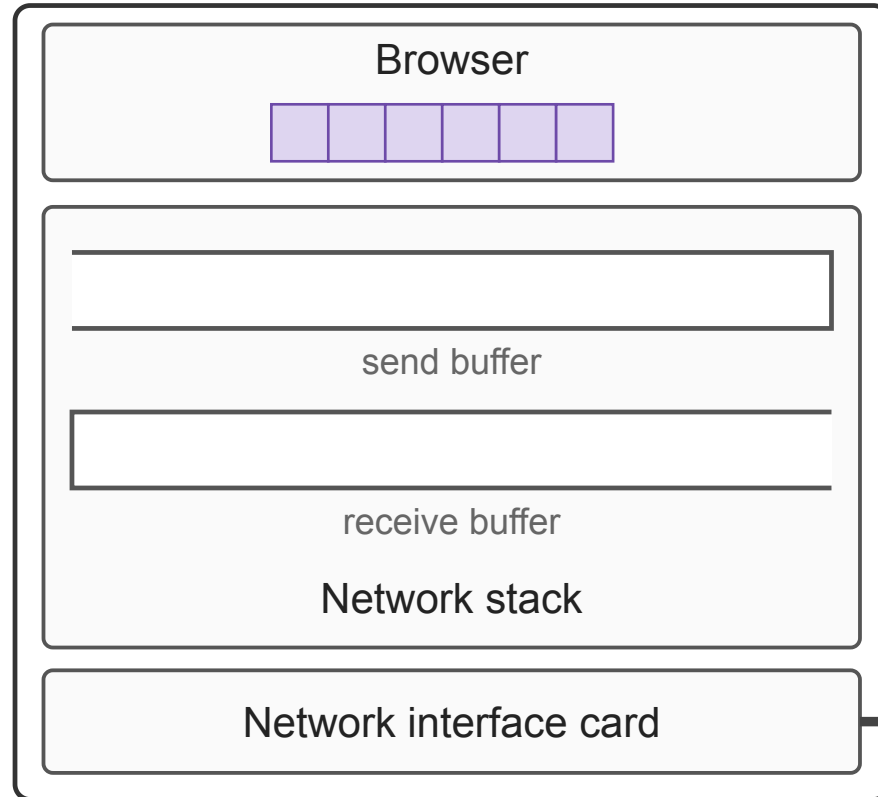


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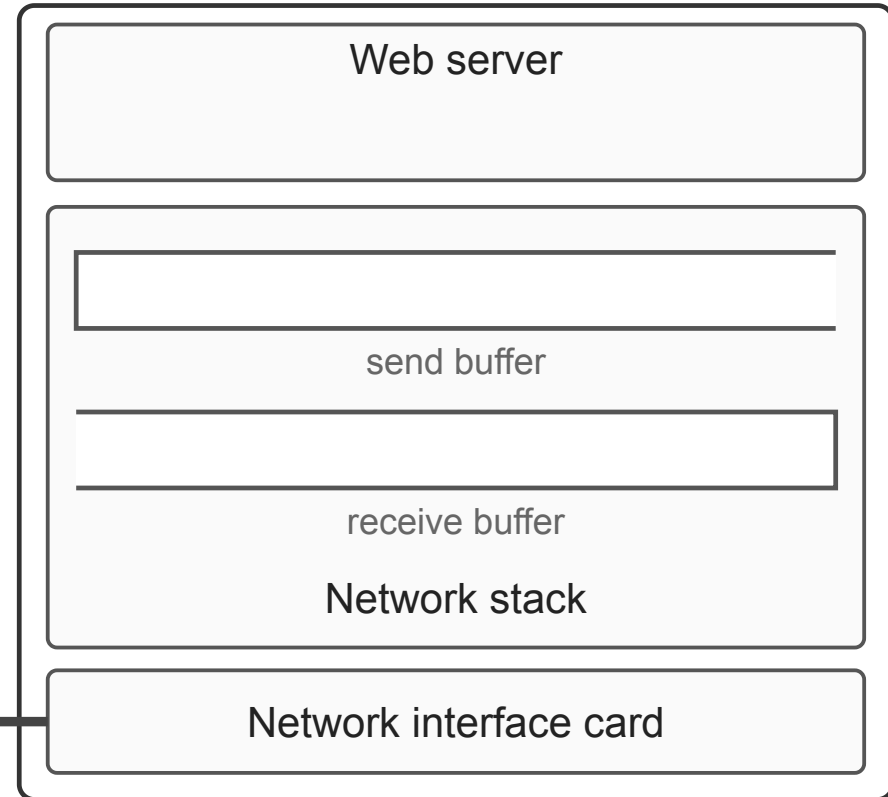


Last time: we built a web browser

A: client



B: server



We also saw

- A page with more than one file, and head-of-line blocking
- Interleaving the files, in the application or in the network stack

Now let's change the application

Outline

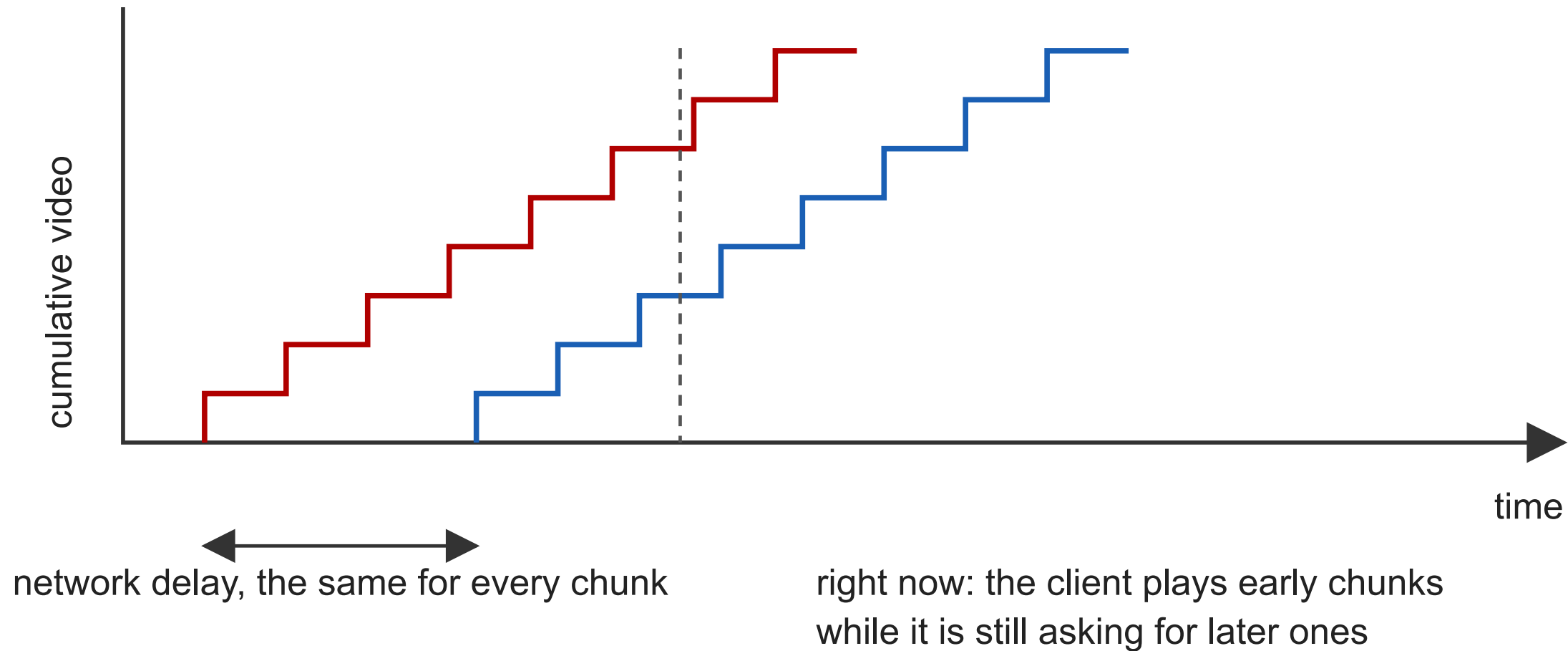
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Video streaming

- A video is also a file on a server that the client can ask for
- But nobody wants to wait for the whole file to download before playing the video
 - hence the term streaming
- Only need to show 30 frames every second
 - not the whole file
- The video is cut into chunks of a few seconds
 - the client asks for the next chunk when it needs it

On a perfect network

- server sends a video chunk in response to the client
- client receives and plays



How would you build the client?

Outline

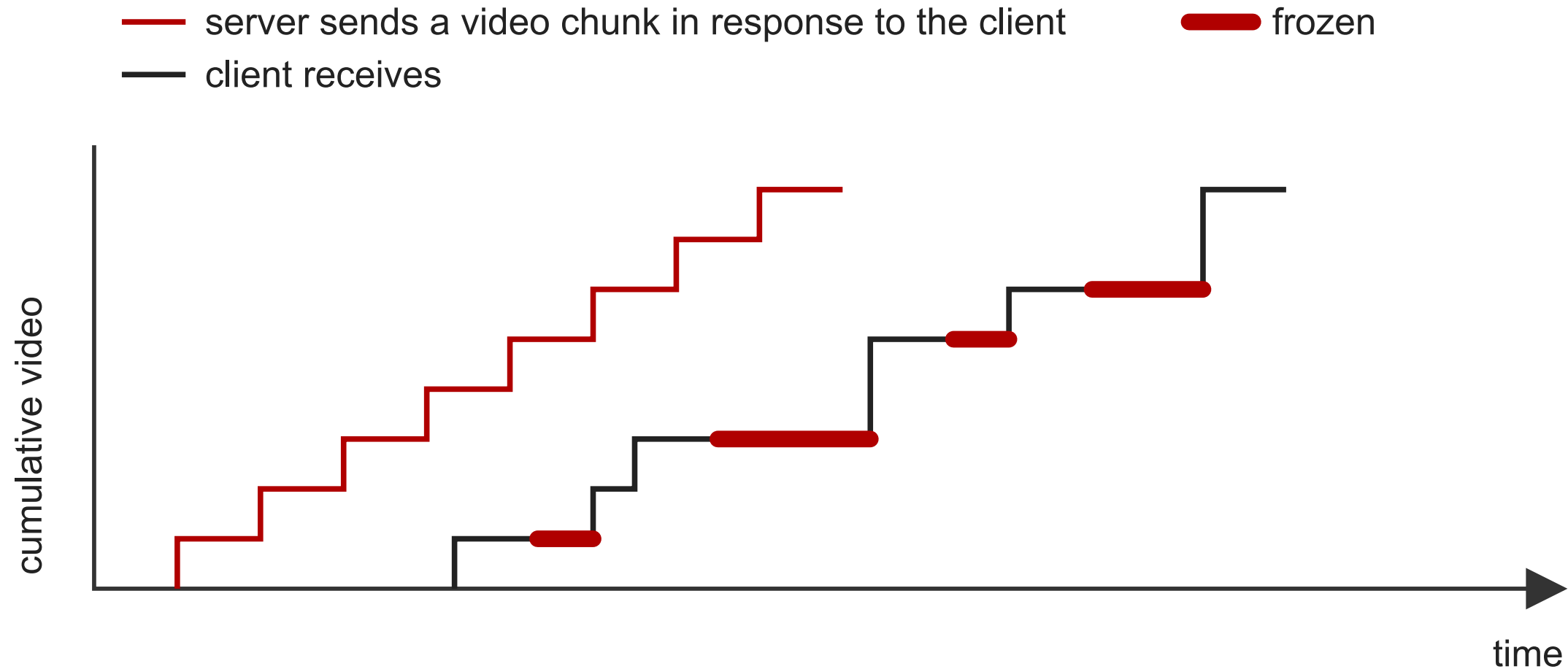
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A real network is not a perfect pipe

- The delay changes from one frame to the next
- Why? Other people share the network (Part 2). For now, take it as given.

How much the delay changes is called jitter

Frames arrive unevenly

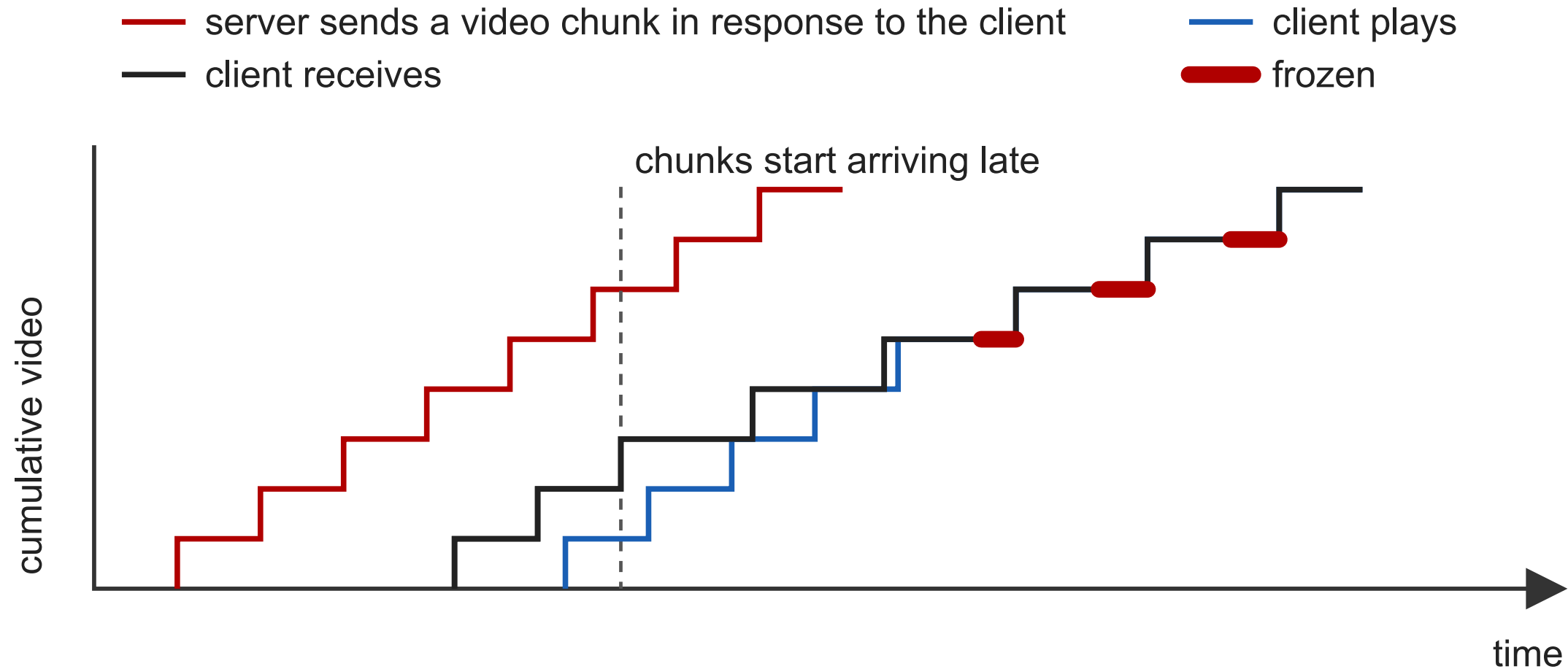


the delay can be different for every chunk

played the moment they arrive: the video keeps freezing

How would you change the client now?

What if chunks keep arriving late?



after the dashed line, every chunk takes longer than 2 s to arrive
the buffer drains, no matter how far ahead the client asks

How would you change the client now?

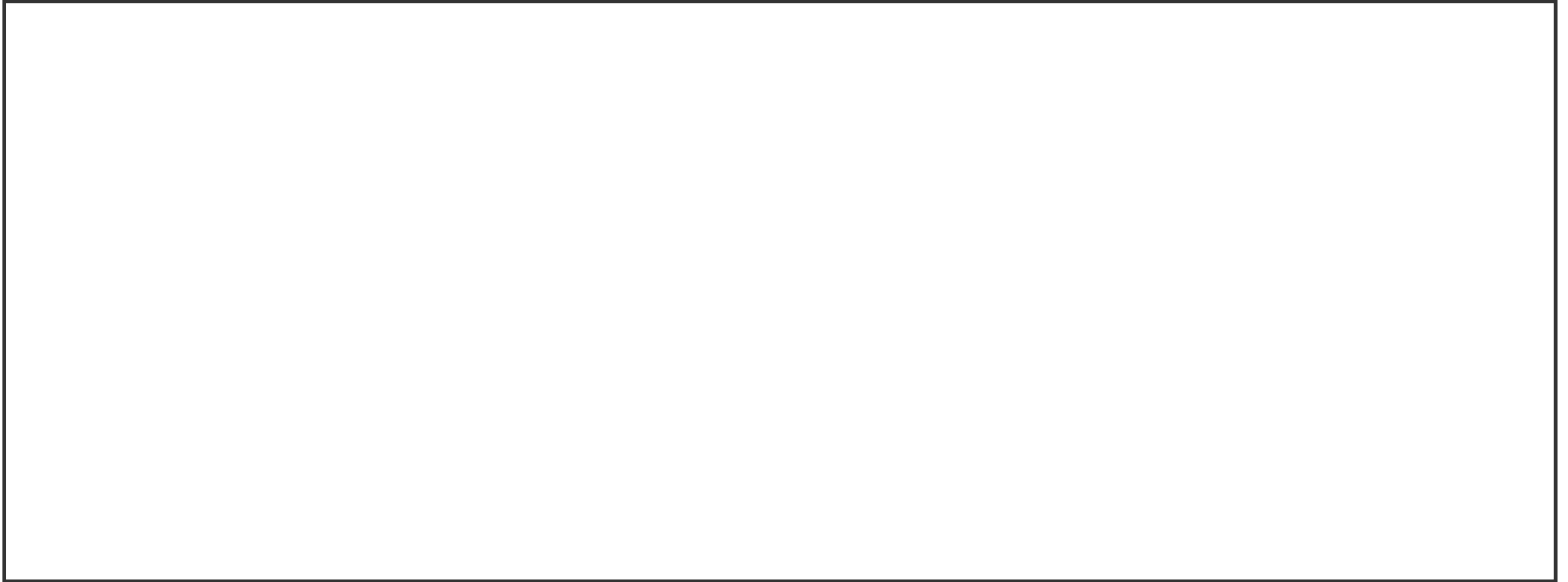
What if you cannot wait at all?

- A video call cannot hold seconds of video in a buffer
- Past about 150 ms of delay each way, people start talking over each other
- So a call wants low delay and low jitter, and would rather skip a late frame than wait for it

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What metrics and properties matter to an application?



What each application wants from the network

	Web browser	Video streaming	Video call